



Reliability modeling of degradation of products with multiple performance characteristics based on gamma processes

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ABSTRACT

Many highly reliable products usually have complex structure, with their reliability being evaluated by two or more performance characteristics. In certain physical situations, the degradation of these performance characteristics would be always positive and strictly increasing. In such a case, the gamma process is usually considered as a degradation process due to its independent and non-negative increments properties. In this paper, we suppose that a product has two dependent performance characteristics and that their degradation can be modeled by gamma processes. For such a bivariate degradation involving two performance characteristics, we propose to use a bivariate Birnbaum–Saunders distribution and its marginal distributions to approximate the reliability function. Inferential method for the corresponding model parameters is then developed. Finally, for an illustration of the proposed model and method, a numerical example about fatigue cracks is discussed and some computational results are presented.

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1. Introduction

For engineering structures and infrastructures, it is generally more attractive to base a failure model on the physical nature of failure and the characteristics of the operating environment [1]. It is recommended, therefore, to model degradation of products in terms of a stochastic process. Markov processes are stochastic processes with independent increments such as the Wiener process with drift, the compound Poisson process, and the gamma process. A Wiener process has found application as a degradation model in many studies due to its good properties. But in certain physical situations, it is often the case that a degradation process should be always positive and strictly increasing. A gamma process is a stochastic process with independent, non-negative increments having a gamma distribution with an identical scale parameter. For the stochastic modeling of monotonic and gradual degradation over time in a sequence of tiny increments, the gamma process is most appropriate for physical factors, such as wearing, corrosion, fatigue crack growth, etc.

In recent years, degradation model based on gamma process has been studied by many authors. Bagdonavicius and Nikulin [2]

modeled degradation of products by a gamma process and included possibly time-dependent covariates. Lawless and Crowder [3] discussed covariates and random effects in a gamma process model with application to degradation and failure. Park and Padgett [4] developed a model for degradation and failure data using a gamma process and incorporated an accelerated test variable. The same authors also discussed the stochastic degradation model based on a gamma process with several accelerating variables [5]. van Noortwijk et al. [6] presented a method to compute the time-dependent reliability of structures where both the deteriorating resistance and the fluctuating load were modeled as stochastic processes. The stochastic process of deterioration was modeled as a gamma process and the stochastic of load was modeled as a combination of a peaks-over-threshold distribution and a Poisson process for the threshold exceedances. Zhou et al. [7] proposed a gamma-based continuous state-space degradation model as an alternative to the Gaussian assumption. Tseng et al. [8] discussed the step-stress accelerated degradation test (SSADT) problem based on gamma process and gave the optimal settings by minimizing the approximate variance of the estimated mean-time-to-failure (MTTF) of the lifetime distribution of the product, and examined the sensitivity and stability of the SSADT plan.

An underlying assumption in all these works is that there is only one single degradation measure. In reality, a system may consist of multiple components or a component having multiple degradation paths, which would make it necessary to consider

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simultaneously multiple degradation measures. For example, a rubidium discharge lamp is the key component of rubidium frequency standard whose degradation in performance can be described by the rubidium consumption or the decreasing intensity of the lamp. In such situations, a bivariate or multivariate degradation model is needed to estimate the reliability of products. This analysis is needed not only for design and technical purposes but also for making managerial decisions.

There are some works on the estimation of system reliability of bivariate or multivariate degradation data. Lu et al. [9] proposed a technique for predicting system performance reliability in real-time considering multiple failure modes. The technique included on-line multivariate monitoring and forecasting of selected performance measures and conditional performance reliability estimates. The performance measures across time were treated as a multivariate time series which were modeled by a state-space approach. Crk [10] assumed that the system failure is governed by several independent mechanisms and presented an effective way to estimate the system's reliability by monitoring each PC. Wang and Coit [11] described a general modeling and analysis approach for reliability prediction based on degradation modeling, considering multiple degradation measures. Xu and Zhao [12] also considered this problem and introduced two methods to model and analyze systems with multiple degradation measures. First, they considered the correlation between the degradation measure and the failure event by introducing a probabilistic measure, and then proposed a state-space model to describe the evolution of the degradation process by incorporating both the degradation dynamics and random stress effects. Barker and Newby [13] took advantage of a degradation model based on multivariate Wiener process to describe the state deterioration of a complex multi-component system and provided an optimal non-periodic inspection policy for it. Savage and Son [14] used the set-theory method to evaluate the structural systems reliability with degrading components. The random variable degradation models and marginal degradation models had been invoked to address the deterioration of structural characteristic, such as material properties and component dimensions. Some other works about multiple failure modes including failure time data are discussed in Huang and Askin [15], Bagdovavicius et al. [16,17], and Belmansour and Nourelfath [18]. Specially, Sari et al. [19] proposed a two-stage reliability model for bivariate degradation data. With the proposed model, not only the marginal reliability but also the system reliability can be assessed. The flexibility of the model to accommodate serial correlation of marginal degradation data, different marginal degradation distribution functions, and dependency between performance characteristics increases the probability to model and analyzes the data more accurately compared with models with stricter assumptions. They analyzed the actual experiment data of the LED tube light system and gave some insights into the failure behavior of the LED lamps system. These works, however, use either independency assumption of the performance characteristics or multivariate normal distribution or based on linear assumption, which are somewhat restrictive in many practical situations.

In this paper, we assume that a product has two performance characteristics whose degradation can be governed by gamma process and take into account the dependency between the performance characteristics. In this regard, we use a bivariate Birnbaum–Saunders distribution [20] and its marginal distributions to approximate the reliability function of the product and develop inferential method for the model parameters.

The rest of the paper is organized as follows. In Section 2, the reliability model of the degradation of products with two performance characteristics based on gamma processes is introduced. In Section 3, method of inference for the model parameters is

presented. A numerical example about fatigue cracks is given in Section 4. Finally, some concluding remarks are made in Section 5.

2. Reliability model of bivariate degradation products

2.1. Bivariate degradation data based on gamma process

In order for the stochastic degradation process to be monotone, we consider it to be a gamma process. A gamma process is a stochastic process with independent, non-negative increments having a Gamma distribution with an identical scale parameter. The gamma process with shape parameter ν and scale parameter u is a continuous time stochastic process $\{X(t), t \geq 0\}$ with the following properties:

- (a) $X(0) = 0$ with probability one;
- (b) $X(t)$ has independent increments;
- (c) $X(t) - X(s) \sim G(\nu(t-s), u)$ for all $t > s \geq 0$,

where $G(x|\nu, u)$ is the probability density function of a random variable X having a gamma distribution with shape parameter ν and scale parameter u given by

$$G(x|\nu, u) = \frac{1}{\Gamma(\nu)u^\nu} x^{\nu-1} \exp(-x/u) I_{(0,\infty)}(x), \quad (1)$$

where

$$I_{(0,\infty)}(x) = \begin{cases} 1 & \text{if } x \in (0, \infty) \\ 0 & \text{if } x \notin (0, \infty) \end{cases}$$

and $\Gamma(\nu) = \int_0^\infty x^{\nu-1} e^{-x} dx$ is the complete gamma function.

In the present work, we suppose that a product has two performance characteristics which are dependent and the performance characteristics are governed by gamma processes. During the degradation test experiment, n items are tested and m measurements for all the items are observed up to the termination time T , which results in degradation measurements $\mathbf{x}_{i,j} = (x_{i,j}^{(1)}, x_{i,j}^{(2)})$ of the i th item corresponding to time t_j , which means that the measurement times of the two performance characteristics and n items are the same (case of balanced data). In general, the bivariate degradation data for this model can be presented in the form:

$$\mathbf{X}_{2n \times m} = \begin{pmatrix} \mathbf{X}^{(1)} \\ \mathbf{X}^{(2)} \end{pmatrix} = \begin{pmatrix} x_{1,1}^{(1)} & \cdots & x_{1,m}^{(1)} \\ \vdots & \ddots & \vdots \\ x_{n,1}^{(1)} & \cdots & x_{n,m}^{(1)} \\ x_{1,1}^{(2)} & \cdots & x_{1,m}^{(2)} \\ \vdots & \ddots & \vdots \\ x_{n,1}^{(2)} & \cdots & x_{n,m}^{(2)} \end{pmatrix}.$$

For $k=1,2$, let

$$\Delta x_{i,j}^{(k)} = x_{i,j+1}^{(k)} - x_{i,j}^{(k)}, \quad \Delta t_j = t_{j+1} - t_j.$$

For each degradation path, by the independent increment property of the gamma process, we have independent but non-identical random variables:

$$\Delta x_{i,j}^{(k)} \sim G(\nu^{(k)} \cdot \Delta t_j, u^{(k)}). \quad (2)$$

So, the probability density function (pdf) of $\Delta x_{i,j}^{(k)}$ is

$$g^{(k)}(\Delta x_{i,j}^{(k)}) = \frac{1}{\Gamma(\nu^{(k)}) \cdot \Delta t_j (u^{(k)})^{\nu^{(k)}}} (\Delta x_{i,j}^{(k)})^{\nu^{(k)} \Delta t_j - 1} \cdot \exp\left(-\frac{\Delta x_{i,j}^{(k)}}{u^{(k)}}\right), \quad (3)$$

with the corresponding mean and variance given by

$$E(\Delta x_{ij}^{(k)}) = v^{(k)} u^{(k)} \Delta t_j \quad \text{and} \quad \text{Var}(\Delta x_{ij}^{(k)}) = v^{(k)} (u^{(k)})^2 \Delta t_j. \quad (4)$$

For the bivariate degradation increments vector $\Delta \mathbf{x}_{ij} = (\Delta x_{ij}^{(1)}, \Delta x_{ij}^{(2)})$, since $\Delta x_{ij}^{(1)}$ and $\Delta x_{ij}^{(2)}$ are dependent, we assume that the correlation coefficient is ρ , and ρ is a constant, which means it is not related to measurement time t , $-1 < \rho < 1$.

For each performance characteristic, we know that the distribution of the first passage time to its threshold value can be approximated by Birnbaum–Saunders distribution [4]. Hence, for the product with two dependent performance characteristics, we utilize a similar idea to approximate the corresponding reliability function.

2.2. Bivariate Birnbaum–Saunders distribution

Birnbaum and Saunders [21] introduced a two-parameter failure time distribution for fatigue failure caused under cyclic loading. The model was developed under the assumption that the failure is due to the development and growth of a dominant crack. Let T be the first passage time of the gamma process X to the threshold C . Park and Padgett [4] obtained the exact distribution of T , and they provided a simpler approximate distribution for T as a two-parameter Birnbaum–Saunders distribution. The cumulative distribution function (cdf) and the corresponding pdf of a two-parameter Birnbaum–Saunders random variable T , for $\alpha > 0$, $\beta > 0$, can be expressed as

$$F_T(t; \alpha, \beta) = \Phi \left[\frac{1}{\alpha} \left(\sqrt{\frac{t}{\beta}} - \sqrt{\frac{\beta}{t}} \right) \right], \quad t > 0, \quad (5)$$

$$f_T(t; \alpha, \beta) = \frac{1}{2\sqrt{2\pi}\alpha\beta} \left[\left(\frac{\beta}{t} \right)^{1/2} + \left(\frac{\beta}{t} \right)^{3/2} \right] \cdot \exp \left[-\frac{1}{2\alpha^2} \left(\frac{t}{\beta} - 2 + \frac{\beta}{t} \right) \right], \quad t > 0. \quad (6)$$

Using a similar transformation on a bivariate normal distribution, Kundu et al. [20] proposed recently the bivariate Birnbaum–Saunders (BVBS) distribution. The bivariate random vector (T_1, T_2) is said to have a bivariate Birnbaum–Saunders distribution with parameters $\alpha_1, \beta_1, \alpha_2, \beta_2, \rho$, if the joint cdf of T_1 and T_2 can be expressed as

$$P(T_1 \leq t_1, T_2 \leq t_2) = \Phi_2 \left[\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right), \frac{1}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right); \rho \right],$$

for $t_1 > 0, t_2 > 0$, and zero otherwise; here, $\alpha_1 > 0, \beta_1 > 0, \alpha_2 > 0, \beta_2 > 0, -1 < \rho < 1$, and $\Phi_2(\cdot, \cdot; \rho)$ is the cdf of a standard bivariate normal vector with correlation coefficient ρ . The corresponding joint pdf of T_1 and T_2 is given by

$$f_{T_1, T_2}(t_1, t_2) = \phi_2 \left[\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right), \frac{1}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right); \rho \right] \times \frac{1}{2\alpha_1\beta_1} \left[\left(\frac{\beta_1}{t_1} \right)^{1/2} + \left(\frac{\beta_1}{t_1} \right)^{3/2} \right] \frac{1}{2\alpha_2\beta_2} \left[\left(\frac{\beta_2}{t_2} \right)^{1/2} + \left(\frac{\beta_2}{t_2} \right)^{3/2} \right],$$

where $\phi_2(\cdot, \cdot; \rho)$ is the pdf of a standard bivariate normal vector with correlation coefficient ρ .

If $(T_1, T_2) \sim BVBS(\alpha_1, \beta_1, \alpha_2, \beta_2, \rho)$, then

(a) $T_i \sim BS(\alpha_i, \beta_i), i = 1, 2$.

(b) The conditional PDF of T_1 , given $T_2 = t_2$, is given by

$$f_{T_1|T_2=t_2}(t_1) = \frac{1}{2\alpha_1\beta_1\sqrt{2\pi(1-\rho^2)}} \left[\left(\frac{\beta_1}{t_1} \right)^{1/2} + \left(\frac{\beta_1}{t_1} \right)^{3/2} \right]$$

$$\times \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right) - \frac{\rho}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right) \right]^2 \right\}.$$

(c) The conditional CDF of T_1 , given $T_2 = t_2$, is given by

$$P[T_1 \leq t_1 | T_2 = t_2] = \Phi \left[\frac{\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right) - \frac{\rho}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right)}{\sqrt{1-\rho^2}} \right];$$

(d) $(T_1^{-1}, T_2^{-1}) \sim BVBS(\alpha_1, \beta_1^{-1}, \alpha_2, \beta_2^{-1}, \rho)$,

$(T_1^{-1}, T_2) \sim BVBS(\alpha_1, \beta_1^{-1}, \alpha_2, \beta_2, \rho)$,

$(T_1, T_2^{-1}) \sim BVBS(\alpha_1, \beta_1, \alpha_2, \beta_2^{-1}, \rho)$.

More properties of the bivariate Birnbaum–Saunders distribution as well as inferential methods have been discussed in detail by Kundu et al. [20].

2.3. Reliability modeling of products with degradation of two performance characteristics

As mentioned above, we suppose that the product has two performance characteristics governed by gamma process and that the two performance characteristics are dependent. Now, we develop an approximation for the reliability function of the product. Let T_1 and T_2 be the first passage times of gamma processes $X^{(1)}$ and $X^{(2)}$ with the threshold values C_1 and C_2 , respectively. Let us assume that the measurement frequency Δt_j is a constant, that is, $\Delta t_j = \Delta t$. From (4), the degradation increment $\Delta x_{ij}^{(k)}$ can be normalized as

$$y_{ij}^{(k)} = \frac{\Delta x_{ij}^{(k)} - v^{(k)} u^{(k)} \Delta t}{\sqrt{v^{(k)} \Delta t u^{(k)}}}.$$

For $\mathbf{y}_{ij} = (y_{ij}^{(1)}, y_{ij}^{(2)})$, it is known that they are i.i.d. vectors satisfying $E(\mathbf{y}_{ij}^{(1)}) = E(\mathbf{y}_{ij}^{(2)}) = 0$, $\text{Var}(\mathbf{y}_{ij}^{(1)}) = \text{Var}(\mathbf{y}_{ij}^{(2)}) = 1$, with correlation coefficient being $\text{corr}(\mathbf{y}_{ij}^{(1)}, \mathbf{y}_{ij}^{(2)}) = \text{corr}(\Delta x_{ij}^{(1)}, \Delta x_{ij}^{(2)}) = \rho$.

Consider the discrete versions of T_1 and T_2 . Let M_1 and M_2 be the discrete first passage times corresponding to the two performance characteristics, respectively. Since the processes $X^{(1)}$ and $X^{(2)}$ are both strictly increasing, the survival probability of the product i after m_1 and m_2 measurements for the two performance characteristics is given by

$$\begin{aligned} P[M_1 > m_1, M_2 > m_2] &= P[X_{M_1}^{(1)} < C_1, X_{M_2}^{(2)} < C_2] \\ &= P \left[\sum_{j=1}^{m_1} \Delta x_{ij}^{(1)} < C_1, \sum_{j=1}^{m_2} \Delta x_{ij}^{(2)} < C_2 \right] \\ &= P \left[\sum_{j=1}^{m_1} \frac{\Delta x_{ij}^{(1)} - v^{(1)} u^{(1)} \Delta t}{\sqrt{v^{(1)} \Delta t u^{(1)}}} < \frac{C_1 - v^{(1)} u^{(1)} m_1 \Delta t}{\sqrt{v^{(1)} \Delta t u^{(1)}}}, \right. \\ &\quad \left. \sum_{j=1}^{m_2} \frac{\Delta x_{ij}^{(2)} - v^{(2)} u^{(2)} \Delta t}{\sqrt{v^{(2)} \Delta t u^{(2)}}} < \frac{C_2 - v^{(2)} u^{(2)} m_2 \Delta t}{\sqrt{v^{(2)} \Delta t u^{(2)}}} \right] \\ &= P \left[\frac{1}{\sqrt{m_1}} \sum_{j=1}^{m_1} y_{ij}^{(1)} < \frac{C_1 - v^{(1)} u^{(1)} m_1 \Delta t}{\sqrt{v^{(1)} m_1 \Delta t u^{(1)}}}, \right. \\ &\quad \left. \frac{1}{\sqrt{m_2}} \sum_{j=1}^{m_2} y_{ij}^{(2)} < \frac{C_2 - v^{(2)} u^{(2)} m_2 \Delta t}{\sqrt{v^{(2)} m_2 \Delta t u^{(2)}}} \right]. \end{aligned} \quad (7)$$

Hunter [22] discussed bivariate central limit theorem, and upon using Lemma 2.2 in that paper, (7) can be approximated by

$$P[M_1 > m_1, M_2 > m_2] \cong \Phi_2 \left(\frac{C_1 - v^{(1)} u^{(1)} m_1 \Delta t}{\sqrt{v^{(1)} m_1 \Delta t u^{(1)}}}, \frac{C_2 - v^{(2)} u^{(2)} m_2 \Delta t}{\sqrt{v^{(2)} m_2 \Delta t u^{(2)}}}; \delta \rho \right), \quad (8)$$

where $\delta = \min\{\sqrt{m_1/m_2}, \sqrt{m_2/m_1}\}$.

Using the properties of standard bivariate normal distribution, (8) can be rewritten as

$$\begin{aligned} P[M_1 > m_1, M_2 > m_2] \\ \cong 1 - \Phi_{M_1} \left(\frac{v^{(1)} u^{(1)} m_1 \Delta t - C_1}{\sqrt{v^{(1)} m_1 \Delta t u^{(1)}}} \right) - \Phi_{M_2} \left(\frac{v^{(2)} u^{(2)} m_2 \Delta t - C_2}{\sqrt{v^{(2)} m_2 \Delta t u^{(2)}}} \right) \\ + \Phi_2 \left(\frac{v^{(1)} u^{(1)} m_1 \Delta t - C_1}{\sqrt{v^{(1)} m_1 \Delta t u^{(1)}}}, \frac{v^{(2)} u^{(2)} m_2 \Delta t - C_2}{\sqrt{v^{(2)} m_2 \Delta t u^{(2)}}}; \delta \rho \right), \end{aligned} \quad (9)$$

where Φ_M is the marginal distribution of $\Phi_2(\cdot, \cdot; \delta \rho)$. A continuous version of M_1 and M_2 can, therefore, be represented as

$$\begin{aligned} P[T_1 > t_1, T_2 > t_2] \\ = 1 - \Phi_{T_1} \left(\frac{v^{(1)} u^{(1)} t_1 - C_1}{\sqrt{v^{(1)} t_1 u^{(1)}}} \right) - \Phi_{T_2} \left(\frac{v^{(2)} u^{(2)} t_2 - C_2}{\sqrt{v^{(2)} t_2 u^{(2)}}} \right) \\ + \Phi_2 \left(\frac{v^{(1)} u^{(1)} t_1 - C_1}{\sqrt{v^{(1)} t_1 u^{(1)}}}, \frac{v^{(2)} u^{(2)} t_2 - C_2}{\sqrt{v^{(2)} t_2 u^{(2)}}}; \delta \rho \right) \\ = 1 - \Phi_{T_1} \left[\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right) \right] - \Phi_{T_2} \left[\frac{1}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right) \right] \\ + \Phi_2 \left[\frac{1}{\alpha_1} \left(\sqrt{\frac{t_1}{\beta_1}} - \sqrt{\frac{\beta_1}{t_1}} \right), \frac{1}{\alpha_2} \left(\sqrt{\frac{t_2}{\beta_2}} - \sqrt{\frac{\beta_2}{t_2}} \right); \delta \rho \right], \end{aligned} \quad (10)$$

where $\delta = \min\{\sqrt{t_1/t_2}, \sqrt{t_2/t_1}\}$, $\alpha_1 = 1/(\sqrt{C_1/u^{(1)}})$, $\beta_1 = C_1/v^{(1)} u^{(1)}$, $\alpha_2 = 1/(\sqrt{C_2/u^{(2)}})$, and $\beta_2 = C_2/v^{(2)} u^{(2)}$. From (10), it is seen that the survival probability of the product can be expressed by the bivariate Birnbaum–Saunders distribution and its marginal distributions.

Now, we consider the reliability of the product at time t (in this case, $t_1 = t_2 = t$). Let

$$U(t) = \frac{1}{\alpha_1} \left(\sqrt{\frac{t}{\beta_1}} - \sqrt{\frac{\beta_1}{t}} \right),$$

$$V(t) = \frac{1}{\alpha_2} \left(\sqrt{\frac{t}{\beta_2}} - \sqrt{\frac{\beta_2}{t}} \right).$$

According to (10), it can be expressed as

$$\begin{aligned} R(t) &= P[T_1 > t, T_2 > t] \\ &= 1 - \Phi_{T_1}[U(t)] - \Phi_{T_2}[V(t)] + \Phi_2[U(t), V(t); \delta \rho]. \end{aligned} \quad (11)$$

So, we can obtain the pdf of the lifetime of the product as follows:

$$\begin{aligned} f(t) &= -\frac{\partial R(t)}{\partial t} = f_{T_1}(t; \alpha_1, \beta_1) \left[1 - \Phi \left(\frac{V(t) - \rho U(t)}{\sqrt{1 - \rho^2}} \right) \right] \\ &+ f_{T_2}(t; \alpha_2, \beta_2) \left[1 - \Phi \left(\frac{U(t) - \rho V(t)}{\sqrt{1 - \rho^2}} \right) \right]. \end{aligned}$$

The details of derivation are presented in Appendix.

If the performance characteristics are independent, the product reliability can be simply written as

$$\begin{aligned} R(t) &= P[T_1 > t] P[T_2 > t] \\ &= 1 - \Phi_{T_1}[U(t)] - \Phi_{T_2}[V(t)] + \Phi_{T_1}[U(t)] \Phi_{T_2}[V(t)]. \end{aligned} \quad (12)$$

3. Parameter estimation

In this section, we discuss the estimation of the model parameters. As described in Section 2.1, let us suppose that n

items are tested and m measurements for all the items are observed up to the termination time T . The measurement frequency of the two performance characteristics are the same and is denoted by Δt for all tested items. For the bivariate degradation increment vector $\Delta \mathbf{x}_{ij}$, we know that the joint distribution of $((1/\sqrt{m-1}) \sum_{j=1}^{m-1} y_{ij}^{(1)}, (1/\sqrt{m-1}) \sum_{j=1}^{m-1} y_{ij}^{(2)})$ can be approximated by standard bivariate normal distribution according to (7) and (8), and $\Delta \mathbf{x}_{ij}^{(k)}$ follows gamma distribution with shape parameter $v^{(k)} \Delta t$ and scale parameter $u^{(k)}$. Thus, the log-likelihood function based on measurements on the two performance characteristics is given by

$$\begin{aligned} L &= \sum_{i=1}^n \log \Phi_2 \left[\frac{1}{\sqrt{m-1}} \sum_{j=1}^{m-1} y_{ij}^{(1)}, \frac{1}{\sqrt{m-1}} \sum_{j=1}^{m-1} y_{ij}^{(2)}; \rho \right] \\ &+ \sum_{k=1}^2 \sum_{i=1}^n \sum_{j=1}^{m-1} \log g^{(k)}(\Delta \mathbf{x}_{ij}^{(k)}). \end{aligned} \quad (13)$$

From (13), we observe that the model is quite complicated from a computational viewpoint. For this reason, we make use of the Bayesian MCMC method for estimating the model parameters. In most practical applications in which Bayesian approach is used, it is difficult to compute analytically the posterior distribution. The MCMC method can be used to generate a sample from the posterior distribution large enough based on a Markov Chain so that any desired feature of the posterior distribution can be accurately estimated. Here, we assume that the prior distributions of all the unknown parameters are non-informative and we then utilize WinBUGS to implement the Gibbs sampling after which we can estimate the model parameters of interest. The details of Gibbs sampling algorithm are presented in [23], for example.

4. Numerical example

In this section, we illustrate the implementation of the model proposed in Section 2. The numerical data are taken from Meeker and Escobar [24], presented in Table C.14, which was reported in Lu and Meeker [25] and read from Fig. 4.52 of Bogdanoff and Kozin [26]. In the original data, 21 samples were tested for fatigue crack development and the measurements are taken at the same measurement times. The time unit is in million cycles. For demonstrating the bivariate degradation, we chose 20 samples and the data are then treated as if half of them form the first performance characteristic and the other half represent the second performance characteristic. It is thus supposed that the product has two possible fatigue crack positions and that this is valid for every unit tested. The data used are the data measured only until 0.09 million cycles. The unit is considered failed if the first crack exceeds 1.6 in. or the second one exceeds 1.3 in. in size, which means $C_1 = 0.7$ and $C_2 = 0.4$. Sari [27] used the data to analyze his models. Now, we also use the same data to illustrate the proposed model. Table 1 lists the crack data. Fig. 1 shows the cumulative degradation of the fatigue crack sizes.

4.1. Estimation of parameters

As described above, there are 10 units and the size of cracks in them is measured each at 0.01 million cycles, which means $n = 10$ and $m = 10$. Let

$$\boldsymbol{\theta} = (v^{(1)}, u^{(1)}, v^{(2)}, u^{(2)}, \rho).$$

Based on the data in Table 1 and the log-likelihood function in (13), we estimate the model parameter $\boldsymbol{\theta}$ by WinBUGS. During the simulation, we generated three Markov chains with different initial values of $\boldsymbol{\theta}$. To ensure that the iterative samples are

independent, we let *thin* to be 15 and generated 50,000 realizations for each chain of θ from the posterior distribution and used the last 120,000 in the estimation of mean, standard deviation, MCMC error and quantiles of the parameters. Table 2 presents the computational results so obtained. Fig. 2 shows the posterior pdf of θ .

It is very important in the use the MCMC algorithm to ensure that the chains converge to the desired posterior distribution if the simulation number is large enough. The Gelman–Rubin method is used to judge the convergence of the chains in WinBUGS. The diagnostic test is implemented by calculating and

Table 1

Fatigue crack data taken from Meeker and Escobar [24].

Cracks	Unit	Millions of cycles									
		0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
Crack A	1	0.90	0.95	1.00	1.05	1.12	1.19	1.27	1.35	1.48	1.64
	2	0.90	0.94	0.98	1.03	1.08	1.14	1.21	1.28	1.37	1.47
	3	0.90	0.94	0.98	1.03	1.08	1.13	1.19	1.26	1.35	1.46
	4	0.90	0.94	0.98	1.03	1.07	1.12	1.19	1.25	1.34	1.43
	5	0.90	0.94	0.98	1.03	1.07	1.12	1.19	1.24	1.34	1.43
	6	0.90	0.94	0.98	1.03	1.07	1.12	1.18	1.23	1.33	1.41
	7	0.90	0.94	0.98	1.02	1.07	1.11	1.17	1.23	1.32	1.41
	8	0.90	0.93	0.97	1.00	1.06	1.11	1.17	1.23	1.30	1.39
	9	0.90	0.92	0.97	1.01	1.05	1.09	1.15	1.21	1.28	1.36
	10	0.90	0.92	0.96	1.00	1.04	1.08	1.13	1.19	1.26	1.34
Crack B	1	0.90	0.90	0.96	1.00	1.04	1.08	1.13	1.18	1.24	1.31
	2	0.90	0.93	0.97	1.00	1.03	1.07	1.10	1.16	1.22	1.29
	3	0.90	0.92	0.97	0.99	1.03	1.06	1.10	1.14	1.20	1.26
	4	0.90	0.93	0.96	1.00	1.03	1.07	1.12	1.16	1.20	1.26
	5	0.90	0.92	0.96	0.99	1.03	1.06	1.10	1.16	1.21	1.27
	6	0.90	0.92	0.95	0.97	1.00	1.03	1.07	1.11	1.16	1.22
	7	0.90	0.93	0.96	0.97	1.00	1.05	1.08	1.11	1.16	1.2
	8	0.90	0.92	0.94	0.97	1.01	1.04	1.07	1.09	1.14	1.19
	9	0.90	0.92	0.94	0.97	0.99	1.02	1.05	1.08	1.12	1.16
	10	0.90	0.92	0.94	0.97	0.99	1.02	1.05	1.08	1.12	1.16

comparing the between-sample (intersample) and the within-sample (intrasample) variability. The details about Gelman–Rubin convergence diagnostic method are presented in [23], for example. The diagnostic results of θ for the three chains are shown in Fig. 3.

From Fig. 3, it is seen that as the number of iterations increase, the intersample (in green) and intrasample (in blue) variability stabilizes, and their ratio (in red) tends to one, which indicates the convergence of the MCMC algorithm.

We also estimated the parameters without considering the dependency between the two fatigue cracks, and these results are presented in Table 3. From Tables 2 and 3, we see that some differences exist in the estimates of the parameters (ν, u) which suggests that the parameters of the two cracks influence each other.

4.2. Simulation analysis

In this paper, we used a bivariate normal distribution to model the data based on a bivariate version of the central limit theorem. In this section, we give a Monte Carlo simulation example to evaluate the accuracy of the proposed approximation. The simulated data sets were generated based on the parameters in Table 2. The algorithms that are capable of generating random vectors possessing any gamma marginal distributions and specified correlation ρ were introduced by Schmeiser and Lal [28]. Let

$$\begin{aligned}\theta &= (\nu^{(1)}, u^{(1)}, \nu^{(2)}, u^{(2)}, \rho) \\ &= (658.9, 9.297e-3, 622.5, 6.152e-3, 0.7298).\end{aligned}$$

Similarly, we set $N=10$, $M=10$, $\Delta t=0.01$ million cycles. According to the model parameter settings above, we simulate 1000 data sets from the model, and then use these data sets for estimating the model parameters. The parameters are estimated by using the Bayesian MCMC method using WinBUGS. 10,000 realizations of the parameters were generated from the posterior

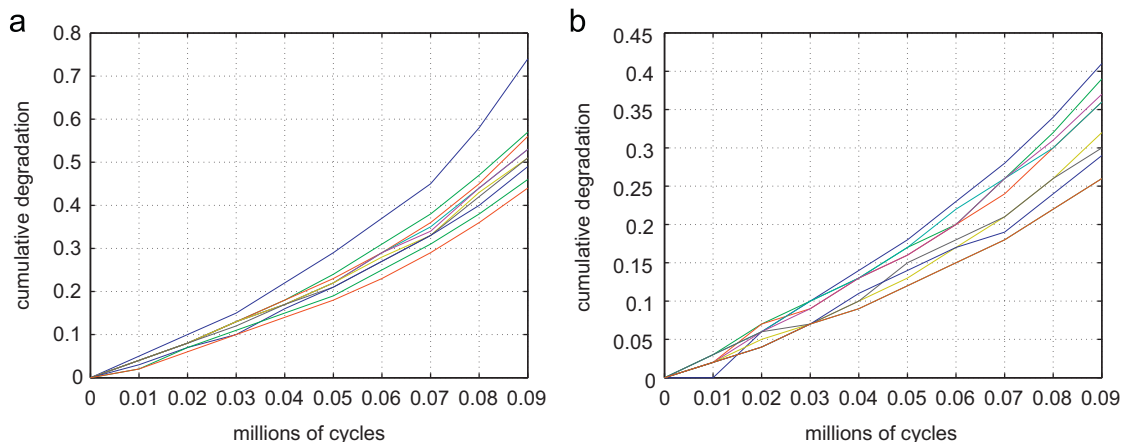


Fig. 1. The cumulative degradation of fatigue crack sizes. (a) Crack A and (b) Crack B.

Table 2

Parameter estimation results considering the dependency.

Parameters	Results					
	Mean	Std	MC-error	2.5%	Median	97.5%
$\nu^{(1)}$	658.9	89.83	0.9761	496.1	654.7	845.2
$u^{(1)}$	$9.297e-3$	$1.316e-3$	$1.400e-5$	$7.075e-3$	$9.178e-3$	0.01220
$\nu^{(2)}$	622.5	87.52	0.9659	464.9	617.8	806.5
$u^{(2)}$	$6.152e-3$	$8.957e-4$	$9.718e-6$	$4.628e-3$	$6.072e-3$	$8.123e-3$
ρ	0.7298	0.1322	$4.207e-4$	0.3890	0.7596	0.8982

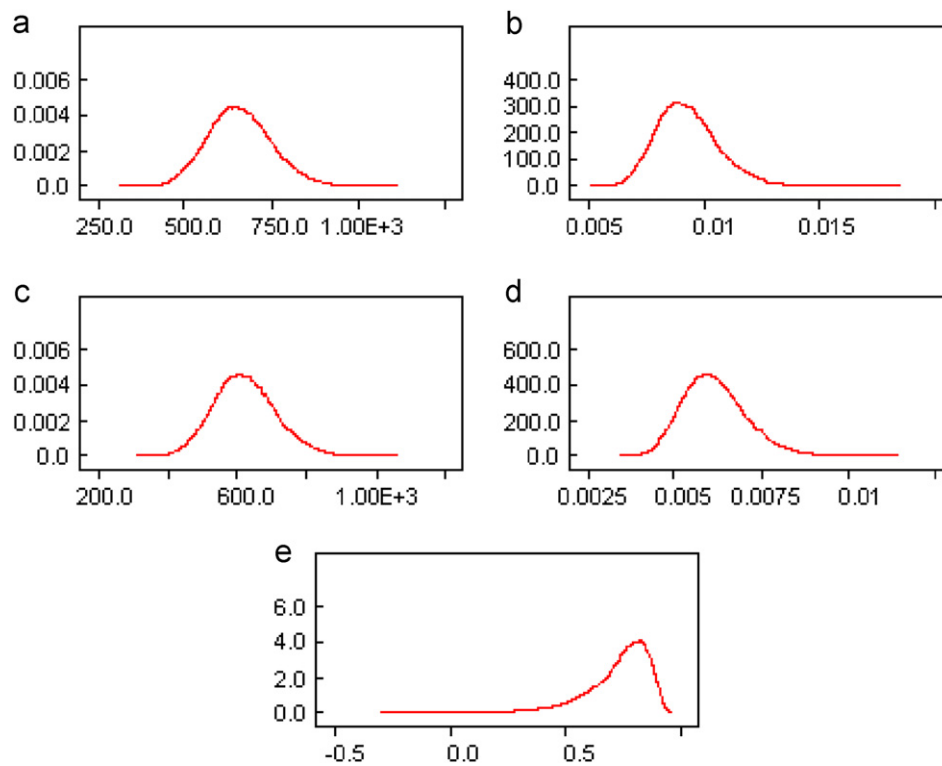


Fig. 2. The posterior pdf of θ . (a) $\nu^{(1)}$, (b) $u^{(1)}$, (c) $\nu^{(2)}$, (d) $u^{(2)}$, (e) ρ .

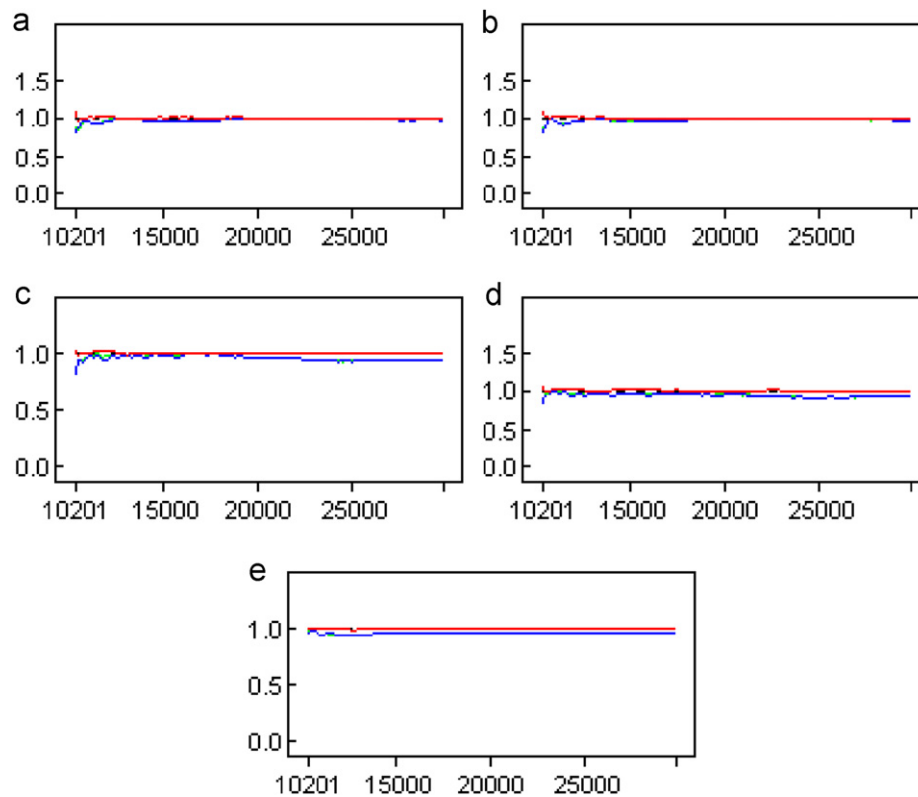


Fig. 3. The Gelman–Rubin convergence diagnostic results of θ . (a) $\nu^{(1)}$, (b) $u^{(1)}$, (c) $\nu^{(2)}$, (d) $u^{(2)}$, (e) ρ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

for each group data. The last 6000 were used for the estimation of mean, standard deviation, MCMC error, and confidence interval of the parameters. Table 4 presents the 95% coverage probability of confidence intervals of θ . We also computed the corresponding

root mean squared error (RMSE) values of these parameters and these results are reported in Table 5. From Table 4, we see that all the coverage probabilities of the parameters θ are over 80% and close to 90%. The entries in Table 5 reveal that RMSE of all the

Table 3

Parameter estimation results without considering the dependency.

Cracks	Parameters	Results					
		Mean	Std	MC-error	2.5%	Median	97.5%
Crack A	ν	733.5	104.9	0.4198	541.3	728.7	952.5
	u	$8.271\text{e}-3$	$1.255\text{e}-3$	$4.997\text{e}-6$	$6.180\text{e}-3$	$8.146\text{e}-3$	0.0111
Crack B	ν	715.7	102.6	0.3734	529.1	711.0	930.8
	u	$5.288\text{e}-3$	$8.062\text{e}-4$	$2.906\text{e}-6$	$3.941\text{e}-3$	$5.208\text{e}-3$	$7.097\text{e}-3$

Table 4

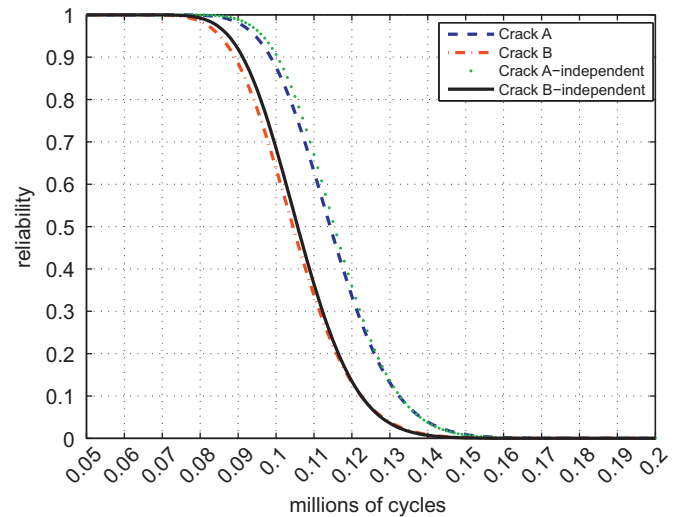
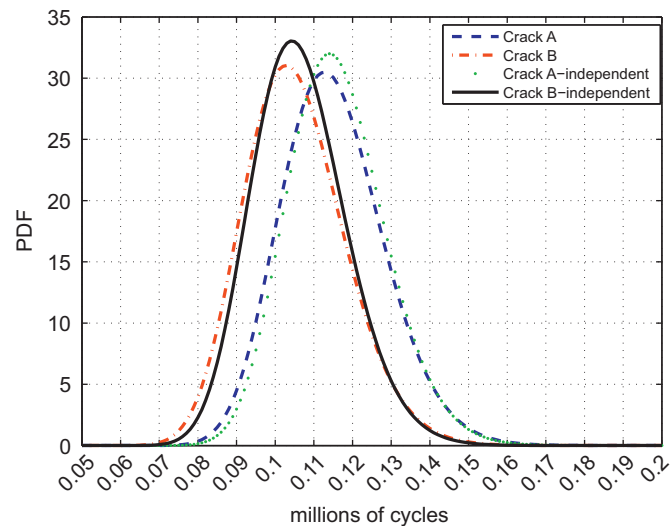
Coverage probabilities of 95% confidence intervals based on Bayesian MCMC simulations.

Parameters	Coverage probability
$\nu^{(1)}$	0.892
$u^{(1)}$	0.849
$\nu^{(2)}$	0.891
$u^{(2)}$	0.824
ρ	0.938

Table 5

Parameter estimation results and corresponding RMSE values.

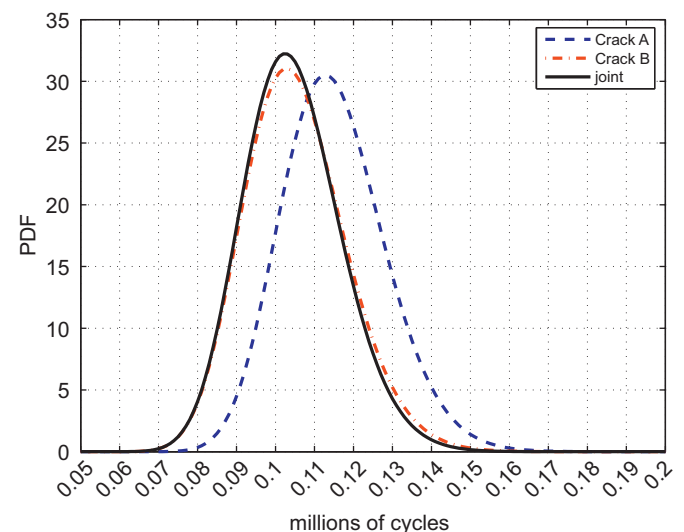
Parameters	True value	Estimated value	RMSE
$\nu^{(1)}$	658.9	631.7	82.30
$u^{(1)}$	$9.297\text{e}-3$	0.01004	$1.382\text{e}-3$
$\nu^{(2)}$	622.5	619.2	73.38
$u^{(2)}$	$6.152\text{e}-3$	$6.627\text{e}-3$	$9.066\text{e}-4$
ρ	0.7298	0.6927	0.0922

**Fig. 5.** Comparison of the product's marginal reliability along with the estimate based on independence assumption.**Fig. 4.** Comparison of the product's marginal pdf along with the estimate based on independence assumption.

parameters are relative small. We, therefore, feel that the proposed model as well as the corresponding inferential method is performing well in this case.

4.3. Reliability assessment

Based on the estimates in Table 2, the marginal and product reliability functions are computed from (11). Product reliability under the independence of the two performance characteristics is

**Fig. 6.** Comparison of the product's marginal pdf and joint pdf.

also obtained according to (12) and by using the estimates of the parameters in Table 3. In Figs. 4 and 5, the marginal pdf and the marginal reliability are presented along with the corresponding estimates based on independence assumption. Fig. 6 compares the marginal pdf and the joint pdf. Fig. 7 shows the product's reliability for both dependent and independent cases. From Figs. 5–7, it is seen that the product's reliability is close to the marginal reliability of Crack B. The reason for this is that the product is considered to have failed if one of the two crack sizes exceeds the threshold value, which means that the two cracks can

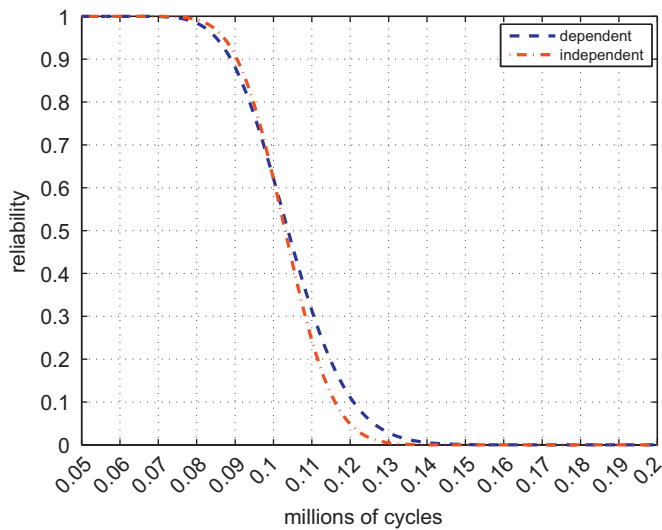


Fig. 7. Comparison of the product's reliability for both dependent and independent cases.

be regarded as a series system, and so the product's reliability should be decided by the crack which exceeds its threshold value at first (which is indeed Crack B).

5. Concluding remarks

In this paper, we have introduced the reliability model for the degradation of products with two performance characteristics by assuming that the performance characteristics are governed by gamma processes. In this case, we use a bivariate Birnbaum-Saunders distribution and its marginal distributions to approximate the reliability of the product. The inference of the model parameters is quite involved due to the complex form of the model and for this reason we used the Bayesian MCMC method for making inference. From the example in Section 4, we find that the proposed model as well as the inferential method for the model parameters works well. Although the development here is for the bivariate case, the reliability model of multivariate degradation of products can be discussed along the same lines. Work is currently under progress in this direction and we hope to report these findings in a future paper.

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Appendix

From (11), we can compute $f(t)$ as follows:

$$f(t) = -\frac{\partial R(t)}{\partial t} = f_{T_1}(t; \alpha_1, \beta_1) + f_{T_2}(t; \alpha_2, \beta_2) - \frac{\partial \Phi_2[U(t), V(t); \rho]}{\partial t},$$

Now, we consider the computation of $\partial \Phi_2[U(t), V(t); \rho] / \partial t$. We know that

$$\Phi_2[U(t), V(t); \rho] = \int_{-\infty}^{V(t)} \int_{-\infty}^{U(t)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(U^2 + V^2 - 2\rho UV)\right\} dU dV.$$

Let

$$p(V, t) = \int_{-\infty}^{U(t)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(U^2 + V^2 - 2\rho UV)\right\} dU,$$

$$p(U, t) = \int_{-\infty}^{V(t)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(U^2 + V^2 - 2\rho UV)\right\} dV.$$

Then, we have

$$\frac{\partial p(V, t)}{\partial t} = \frac{\partial U(t)}{\partial t} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(U(t)^2 + V^2 - 2\rho U(t)V)\right\}.$$

Also,

$$\begin{aligned} \frac{\partial \Phi_2[U(t), V(t); \rho]}{\partial t} &= \frac{\partial V(t)}{\partial t} p(V(t), t) + \int_{-\infty}^{V(t)} \frac{\partial p(V, t)}{\partial t} dV \\ &= \frac{\partial V(t)}{\partial t} p(V(t), t) + \frac{\partial U(t)}{\partial t} p(U(t), t), \end{aligned}$$

where

$$\begin{aligned} p(V(t), t) &= \int_{-\infty}^{U(t)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(U^2 + V(t)^2 - 2\rho UV(t))\right\} dU \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{V(t)^2}{2}\right) \int_{-\infty}^{U(t)} \frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left\{-\frac{(U - \rho V(t))^2}{2(1-\rho^2)}\right\} dU \\ &= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{V(t)^2}{2}\right) \Phi\left(\frac{U(t) - \rho V(t)}{\sqrt{1-\rho^2}}\right); \end{aligned}$$

Similarly, we obtain

$$p(U(t), t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{U(t)^2}{2}\right) \Phi\left(\frac{V(t) - \rho U(t)}{\sqrt{1-\rho^2}}\right).$$

Thus,

$$\begin{aligned} \frac{\partial \Phi_2[U(t), V(t); \rho]}{\partial t} &= \frac{\partial V(t)}{\partial t} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{V(t)^2}{2}\right) \Phi\left(\frac{U(t) - \rho V(t)}{\sqrt{1-\rho^2}}\right) \\ &\quad + \frac{\partial U(t)}{\partial t} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{U(t)^2}{2}\right) \Phi\left(\frac{V(t) - \rho U(t)}{\sqrt{1-\rho^2}}\right) \\ &= f_{T_2}(t; \alpha_2, \beta_2) \Phi\left(\frac{U(t) - \rho V(t)}{\sqrt{1-\rho^2}}\right) \\ &\quad + f_{T_1}(t; \alpha_1, \beta_1) \Phi\left(\frac{V(t) - \rho U(t)}{\sqrt{1-\rho^2}}\right), \end{aligned}$$

and so

$$f(t) = f_{T_1}(t; \alpha_1, \beta_1) \left[1 - \Phi\left(\frac{V(t) - \rho U(t)}{\sqrt{1-\rho^2}}\right)\right] + f_{T_2}(t; \alpha_2, \beta_2) \left[1 - \Phi\left(\frac{U(t) - \rho V(t)}{\sqrt{1-\rho^2}}\right)\right].$$

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